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Why do workload spikes cause injuries, and which athletes are at higher risk? Mediators and moderators in workload–injury investigations

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Spikes in training and competition workloads, especially in undertrained athletes, increase injury risk.[1] However, just as attributing athletic injuries to single risk factors is an oversimplification of the injury process,[2 3] interpreting this workload-injury relationship should not be done in isolation. Instead, we must further unpack how (ie, through which mechanisms) workload spikes might result in injury, and what characteristics make athletes more robust or more susceptible to injury at any given workload. In other words, which factors *mediate* the workload-injury relationship, and which *moderate* the relationship.

DOMINO OR DIMMER? DIFFERENTIATING 'MEDIATORS' AND 'MODERATORS'

Like dominoes being knocked over, *mediators* can be viewed as the intermediary steps that explain the association between an observed variable and an outcome.[4] In this context, mediating variables help to explain 'why changes in workloads might cause injuries?' For example, it is known that rugby league players exposed to spikes in running workloads, indicated by a high acute:chronic workload ratio, are at an increased risk for non-contact injuries.[5] One potential explanation is that neuromuscular fatigue mediates this relationship, such that increased workloads cause higher levels of neuromuscular fatigue, subsequently precipitating injury (eg, while performing a cutting manoeuvre) (figure 1).

On the other hand, *moderators* can be viewed as 'dimmer switches', modifying the effect of a given variable on an outcome. They are also referred to as *interactions*, or *effect modifiers*.[6] In our context, moderator variables answer the question: 'what characteristics make certain athletes more robust or more susceptible to injury at given workloads'? For example, in Gaelic football, high aerobic fitness protects against workload spikes.[7] In other words, aerobic fitness protectively moderates the workload effect by 'dimming' or reducing the risk of rapid workload increases (figure 1).

NOT SO SIMPLE, NOT SO FAST

Admittedly, these previous examples are simple illustrations of how mediation and moderation are conceptualised. In reality, the aetiology of injury is complex, dynamic, multifactorial and context dependent.[2 3] Therefore, for certain injuries, the effect of workload on injury is most likely more appropriately viewed as moderated mediation.[4] Combining the two aforementioned examples, a spike in workload may produce increased levels of neuromuscular fatigue, but the strength of that

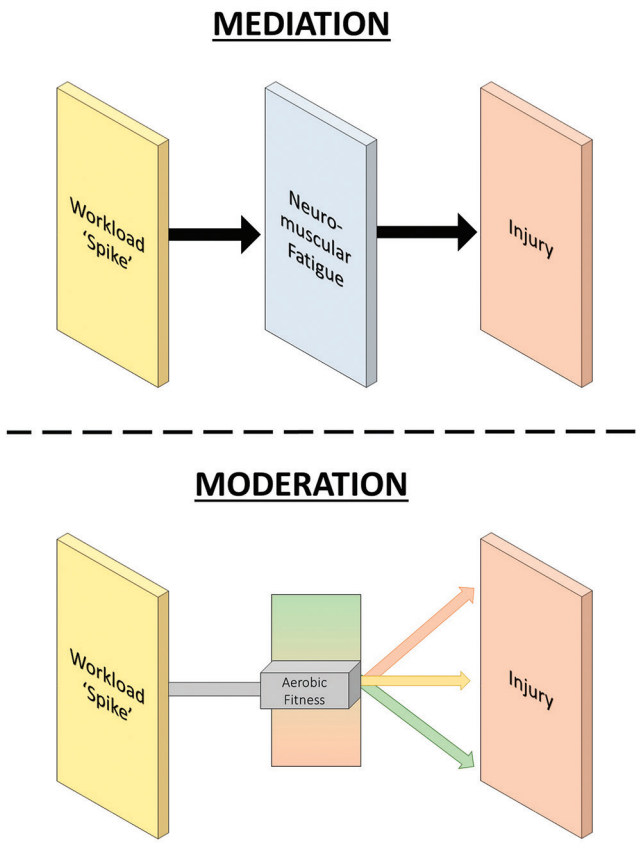

Figure 1 Examples of how mediating and moderating variables explain the association between a workload spike and subsequent injury. Top, neuromuscular fatigue is shown as a mediator (ie, domino) of the effect of workload spikes on injury. In this example, a spike in workload causes an increase in neuromuscular fatigue, and an increase in neuromuscular fatigue leads to an injury. Bottom, aerobic fitness acts as a moderator (ie, dimmer switch) of the relationship between workload spikes and injury. In this case, a given spike in workload will elicit a different injury risk dependent on an individual's fitness level.

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relationship may be moderated by aerobic fitness. In complex systems language, this example may manifest in a 'risk profile', which includes workload, aerobic fitness and neuromuscular fatigue as interacting factors within a 'web of determinants'—related to an 'emergent pattern' of non-contact injuries.[2]

Expounding the causal relationship of workloads and injuries and explaining more complex causal pathways present both methodological and analytical hurdles. First, study design and data collection must be conducted in a way so that mediator and moderator variables are both considered and measured. In reducing the plausibility of alternative explanations, randomised experimental designs should be considered where possible.[4] The a priori selection and measurement of potential confounding variables, which do not lie on the causal pathway, should be simultaneously conducted.[6] Incorporating these additional variables requires greater sample sizes (ie, increased power) and a large number of injuries to demonstrate these mediation/moderation effects.[4] Finally, adequate analysis of mediation and moderation often requires both complex designs, and more complex analyses, including structural equation modelling and multilevel modelling, to name a few.[4]

WHAT REWARDS AWAIT AT THE FINISH LINE AFTER JUMPING THE HURDLES?

1. *Improved understanding, better prevention*—Mediation and moderation, as well as their more complicated cases, will improve our understanding of causal mechanisms behind injuries, and subsequent preventative strategies. Moderating variables that are modifiable are key intervention points to 'dim' the risk associated with training and competition workloads.
2. *Addressing insufficiencies*—The inability of screening single risk factors to predict future injuries has been understandably challenged.[8] However, shifting our conceptualisation of these variables from injury predictors to variables that moderate the load-injury relationship provides a different motivation for screening and addressing any shortcomings. For example, while aerobic fitness may not predict injuries, it may increase athletes' resilience to higher workloads. Therefore, when athletes perform poorly on preseason fitness tests, practitioners may choose to lower the allowable workload 'threshold' for this athlete, while providing individualised attention to address the deficiency.
3. *Training smarter and harder*—Ultimately, attaining high chronic training loads without rapid spikes in the process is considered current best practice.[1] However, understanding the characteristics that make certain athletes more robust may allow for more nuanced training load prescriptions. For athletes with a collection of characteristics that 'dim' workload-related injury risks (eg, high aerobic fitness, no previous injuries), practitioners may consider prescribing higher training workloads (eg, an acute:chronic workload ratio of 1.7) for performance purposes, given that their associated risk is lower than that of the average athlete.

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